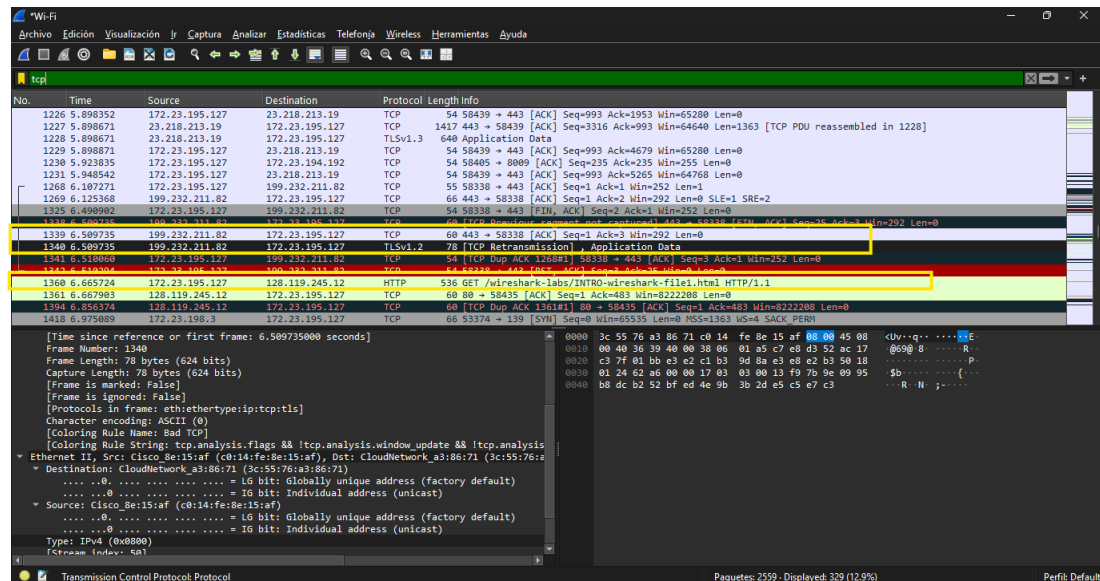


Wireshark Lab: Getting Started v9.0

Jakes Quilumbaquin De La Cruz

Now answer the questions below. If you're doing this lab as part of class, your teacher will provide details about how to hand in assignments, whether written or in a learning management system (LMS).¹ If you're unable to run Wireshark on a live network connection or are answering questions via an LMS, you can download a packet trace file that was captured while following the steps above².

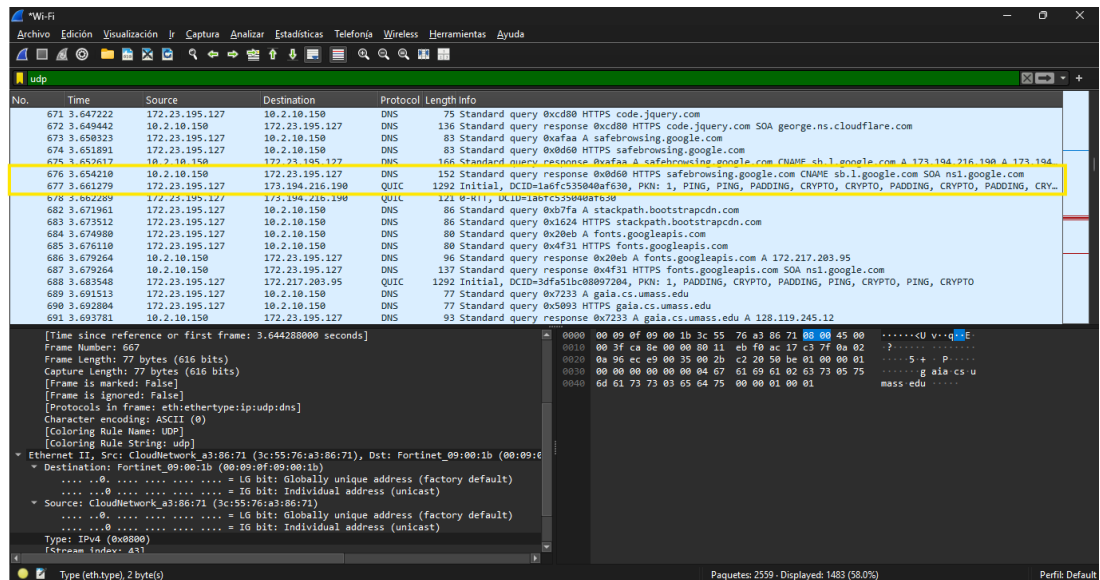
1. Which of the following protocols are shown as appearing (i.e., are listed in the Wireshark "protocol" column) in your trace file: TCP, QUIC, HTTP, DNS, UDP, TLSv1.2?



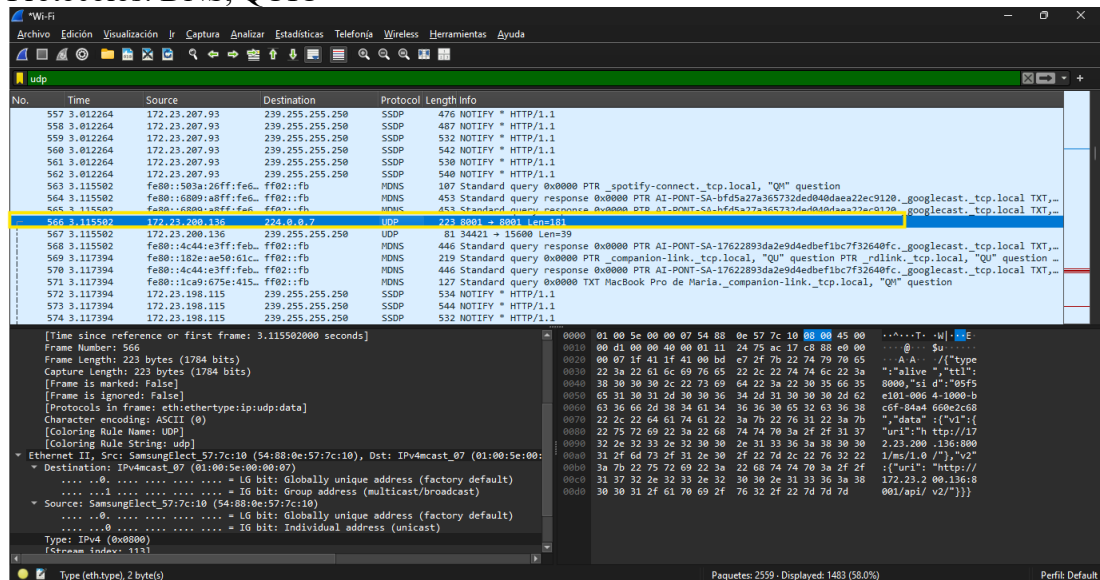
Protocolos: TCP, HTTP, TLSv1.2

¹ For the author's class and written answers, students print out the GET and response messages and indicate where in the message they've found the information that answers a question. They do this by marking paper copies with a pen or annotating electronic copies with text in a colored font. There are LMS modules for teachers that allow students to answer these questions online and have answers auto-graded for these Wireshark labs at http://gaia.cs.umass.edu/kurose_ross/lms.htm

² You can download the zip file <http://gaia.cs.umass.edu/wireshark-labs/wireshark-traces-9e.zip> and extract the trace file into-wireshark-trace1. This trace file can be used to answer these Wireshark lab questions without actually capturing packets on your own. Each trace was made using Wireshark running on one of the author's computers, while performing the steps indicated in the Wireshark lab. Once you've downloaded a trace file, you can load it into Wireshark and view the trace using the *File* pull down menu, choosing *Open*, and then selecting the trace file name.



Protocolos: DNS, QUIC



Protocolos: UDP

- How long did it take from when the HTTP GET message was sent until the HTTP OK reply was received? (By default, the value of the Time column in the packet-listing window is the amount of time, in seconds, since Wireshark tracing began. (If you want to display the Time field in time-of-day format, select the Wireshark *View* pull down menu, then select *Time Display Format*, then select *Time-of-day*.)

No.	Time	Source	Destination	Protocol	Length	Info
1360	6.665724	172.23.195.127	128.119.245.12	HTTP	536	GET /wireshark-labs/INTRO-wireshark-file1.html HTTP/1.1
1461	7.161694	128.119.245.12	172.23.195.127	HTTP	495	HTTP/1.1 200 OK (text/html)
1487	7.279521	172.23.195.127	128.119.245.12	HTTP	482	GET /favicon.ico HTTP/1.1
1541	7.493280	128.119.245.12	172.23.195.127	HTTP	636	HTTP/1.1 301 Moved Permanently (text/html)

HTTP GET : 6.665724

HTTP OK: 7.161694

Tiempo transcurrido: OK – GET = 0.49597

- What is the Internet address of the gaia.cs.umass.edu (also known as www-net.cs.umass.edu)? What is the Internet address of your computer or (if you are using the trace file) the computer that sent the HTTP GET message?

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The internet address of the gaia.cs.umass.edu is 128.119.245.12 and of my computer 172.23.195.127

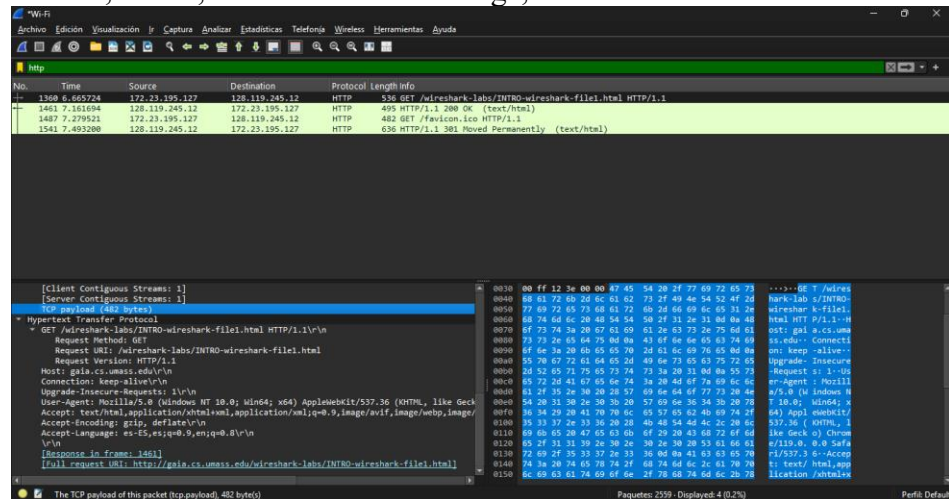
To answer the following two questions, you'll need to select the TCP packet containing the HTTP GET request (hint: this is packet number 286³). The purpose of these next two questions is to familiarize you with using Wireshark's "Details of selected packet window"; see Figure 3. To do this, click on Packet 286 (your screen should look similar to Figure 3). To answer the first question below, then look in the "Details of selected packet" window toggle the triangle for HTTP (your screen should then look similar to

³ Remember that this "packet number" is assigned by Wireshark for listing purposes only; it is NOT a packet number contained in any real packet header.

Figure 5); for the second question below, you'll need to expand the information on the Transmission Control Protocol (TCP) part of this packet.

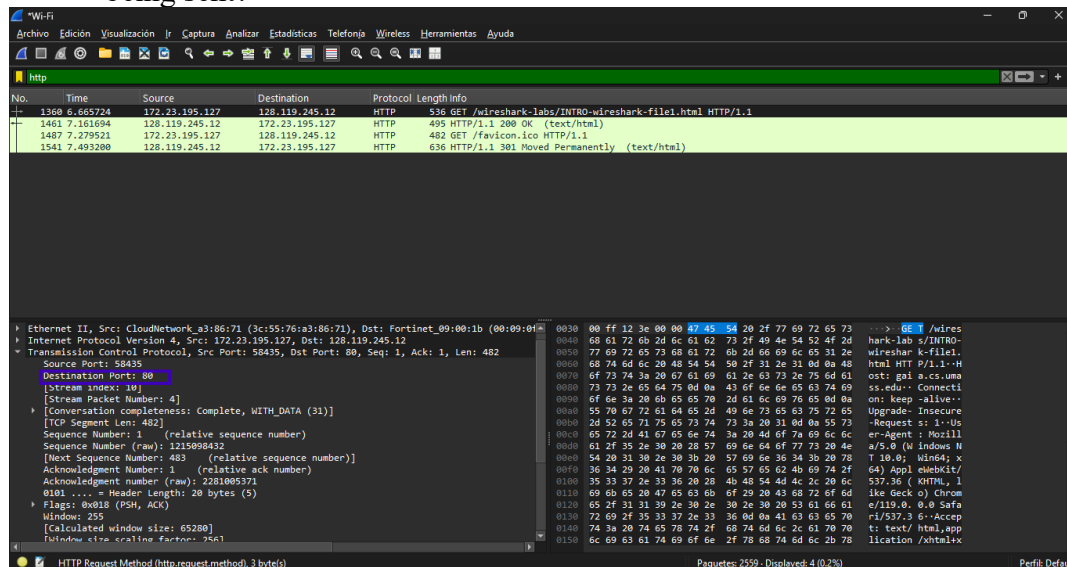
- Expand the information on the HTTP message in the Wireshark "Details of selected packet" window (see Figure 3 above) so you can see the fields in the HTTP GET request message. What type of Web browser issued the HTTP request? The answer is shown at the right end of the information following the "User-Agent:" field in the expanded HTTP message display. [This field value in the HTTP message is how a web server learns what type of browser you are using.]

- Firefox, Safari, Microsoft Internet Edge, Other



Chrome

- Expand the information on the Transmission Control Protocol for this packet in the Wireshark "Details of selected packet" window (see Figure 3 in the lab writeup) so you can see the fields in the TCP segment carrying the HTTP message. What is the destination port number (the number following "Dest Port:." for the TCP segment containing the HTTP request) to which this HTTP request is being sent?



Destination Port: 80
And finally ...

6. Print the two HTTP messages (GET and OK) referred to in question 2 above. To do so, select *Print* from the Wireshark *File* command menu, and select the “*Selected Packet Only*” and “*Print as displayed*” radial buttons, and then click OK.

